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JDBC (PART 2)

8.6 JDBC Components :

- **Driver (Translator) :-** It is used to convert java calls to database specific calls and database specific calls to java calls
- **Connection (Road) :-** TO communicate java with database
- **Statement (Truck) :-** To send SQL Query from java to database and to get the result
- **ResultSet (Box) :-** Output of SQL query

8.7 Steps to develop JDBC Application :

1. Load & Register driver
2. Establish connection b/w java application and database
3. Create statement object
4. Send and execute SQL query
5. Process result in the form of ResultSet
6. Close all the resources (connection, statement, resultset)

Coding Part

```
import java.sql.*;


Class.forName("DRIVER CLASS NAME");

Connection con =
DriverManager.getConnection("JDBC
URL/CONNECTION URL", "USERNAME",
"PASSWORD");

Statement st=con.createStatement();

ResultSet rs=st.executeQuery("SQL QUERY");

while(rs.next)
{
    //body to get values from database
}
con.close();

st.close();

rs.close();
```

What we required for each database ??????

1. DRIVER CLASS NAME
2. JDBC URL/CONNECTION URL
3. USERNAME
4. PASSWORD
5. DRIVER SOFTWARE (jar file)

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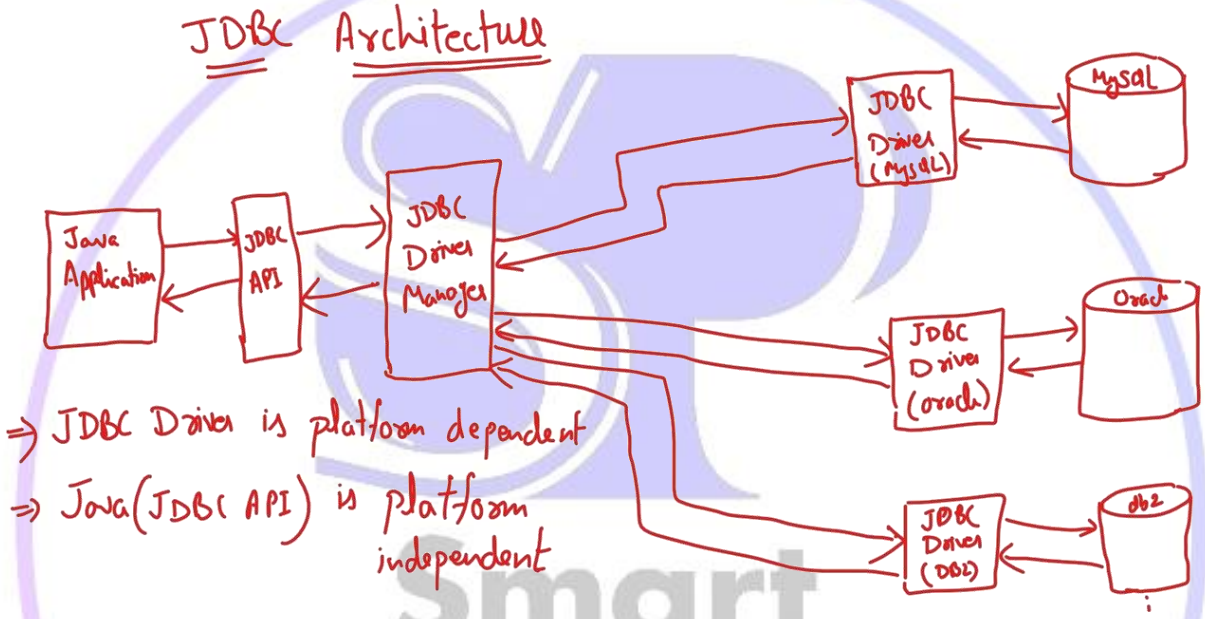
8. JDBC Architecture :

Diagram



1. JDBC Driver is platform dependent
2. Java is platform independent

8.9 What is Driver ?

- Driver is an interface provided by Sun Microsystems and implemented by database vendors
- Driver is used like translator between java application and database which converts the java specific calls into database specific calls and vice versa

- **DRIVER SOFTWARE :** It is just a jar file which contains implementation part provided by different database vendors

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8.10 TYPES OF DRIVER ?

1. Type 1 Driver (Bridge Driver) or (JDBC-ODBC Bridge Driver)
2. Type 2 Driver (Native Driver) or (Native API Partly Java Driver)

3. Type 3 Driver (Middleware Driver) or (All Java Net Protocol Driver) (Network Protocol Driver)
4. Type 4 Driver (Pure Java Driver) (Thin Driver) or (All Java Native Protocol Driver)

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